Explainable Multimodal AI for Recognition and Interpretation of Indian Temple Architectural Heritage

Complete Research Project Documentation

Indian Knowledge Systems (IKS) — Unit IV: Town Planning & Architecture / Temple Architecture

Proposed project for an undergraduate CSE research-oriented project

Project focus: Computer Vision + Deep Learning + Explainable AI + Indian Architectural Knowledge + Digital Heritage

Recommended research direction: A feature-aware system that identifies temple architectural traditions, detects meaningful architectural components, and provides evidence-grounded explanations instead of relying only on whole-image classification.

1. Executive Summary

Indian temple architecture represents a highly structured body of architectural knowledge involving spatial organization, geometry, construction, ornamentation and regional traditions. This project proposes an Explainable Multimodal Artificial Intelligence framework for automated recognition and interpretation of Indian temple architectural heritage. Rather than producing only a style label, the system will identify architecturally meaningful visual features such as shikhara/vimana forms, gopurams, mandapas, pillars, bases and ornamental structures, and use these features to support classification.

The central research idea is to combine whole-image visual representations with architectural component information and a structured IKS knowledge layer. Explainable AI methods such as Grad-CAM and feature attribution will show which regions and architectural components influenced a prediction. A temple-disjoint evaluation protocol will be used so that photographs of the same temple do not appear in both training and test sets.

```
Temple Image
↓
Image Preprocessing
↓
Architectural Component Detection
↓
Feature Extraction
↓
Feature Fusion
↓
Style Classification
↓
Explainable AI
↓
IKS Knowledge Mapping
↓
Human-readable Architectural Explanation
```

2. Proposed Research Title

Primary title: Explainable Multimodal AI for Recognition and Interpretation of Indian Temple Architectural Heritage

Alternative title: A Feature-Aware and Explainable Deep Learning Framework for Automated Classification of Indian Temple Architectural Heritage

3. IKS Syllabus Alignment

Syllabus component	Project connection
Unit IV — Indian Architecture	Core domain: temple architecture and architectural traditions.
V■stu ■■stra	Used as part of the structured architectural knowledge layer where appropriate.
Temple Architecture	Primary knowledge domain and annotation taxonomy.
Town Planning & Architecture	Contextual analysis of spatial and structural features.
Unit II — Linguistics / Knowledge Frameworks	Optional textual knowledge layer and provenance-aware representation.
Unit III — Science, Engineering & Technology	Computer vision, computational analysis and digital preservation.

4. Why This Topic Is Important

- Indian temples contain diverse structural, geometric and decorative features that can be digitally documented and analyzed.
- Manual architectural identification requires domain expertise and can be time-consuming.

- Computer vision can support heritage documentation, education, digital archives and virtual heritage applications.
- Explainability is important because a heritage-analysis system should show architectural evidence for a prediction rather than only output a class label.
- The project provides a direct bridge between an IKS knowledge domain and modern CSE research methods.

5. Existing Research and the Need for a New Contribution

Recent research has already demonstrated deep-learning classification of Indian temple architectural styles. A 2026 study reported a dataset of 9,046 temple images from 218 temples and compared EfficientNet-B0, MobileNetV2, MobileNetV3, GhostNet and ShuffleNetV2 for N■gara, Vesara and Dr■vi■a classification. The study also used Grad-CAM for interpretation.

Therefore, a project framed simply as “CNN-based classification of Indian temple architecture” would largely reproduce an existing research direction. The proposed work instead focuses on architectural component detection, feature-aware classification, explanation alignment and knowledge-grounded interpretation.

> **Proposed research gap:** Existing work demonstrates that deep-learning models can classify Indian temple images, but there remains scope for a feature-aware system that explicitly detects architectural components, evaluates whether explanations correspond to those components, and connects visual predictions to structured IKS architectural knowledge.

6. Research Problem

The project addresses four related problems:

- **Black-box classification:** A model may produce a correct label without showing which architectural evidence it used.
- **Whole-image bias:** Background, vegetation, sky, crowds or image-source characteristics can become unintended predictive cues.
- **Architectural complexity:** Indian temple traditions contain regional and historical variations, so rigid categories can oversimplify the domain.
- **Lack of architectural explanation:** A useful heritage system should answer why a classification was made in terms meaningful to an architectural researcher.

7. Central Research Question

Can a feature-aware and explainable deep-learning system identify Indian temple architectural traditions using architecturally meaningful visual characteristics rather than relying primarily on image-level or contextual correlations?

8. Secondary Research Questions

- Can deep-learning models distinguish selected architectural traditions from previously unseen temples?
- Which architectural components contribute most strongly to classification?
- Does explicit architectural component detection improve classification compared with a conventional whole-image CNN?
- Can Explainable AI outputs be aligned with documented architectural features?
- Does combining visual evidence with structured architectural knowledge improve interpretability?

9. Research Hypotheses

- A feature-aware model incorporating architectural component information will provide more interpretable predictions than a conventional image-level classifier.
- Explicit architectural feature detection will reduce dependence on irrelevant contextual regions.
- Combining visual classification with structured architectural knowledge will improve explanation quality.

10. IKS and Architectural Knowledge Foundation

The IKS layer should be a real part of the research methodology rather than a historical introduction added only to satisfy the syllabus. The system should build a structured taxonomy of architectural components and link each component to reliable sources, definitions, visual characteristics and example monuments.

Feature	Role in the project	Possible visual characteristics
■ikhara	Northern-style superstructure analysis	Curvilinear/vertical tower forms in relevant traditions
Vim■na	Southern temple superstructure analysis	Stepped/pyramidal composition in relevant examples
Gopura/Gopuram	Gateway detection	Monumental gateway structures
Ma■■apa	Pillared-hall detection	Pillared halls and roof structures
Jagati / base	Raised-platform analysis	Elevated plinth/platform
Amalaka	Crowning-element analysis	Ribbed circular crowning form in relevant examples
Pillars	Structural/decorative feature	Repeated vertical structural elements

11. Scholarly Classification Caveat

The paper should not claim that every Indian temple belongs cleanly to exactly one of N■gara, Dr■vi■a or Vesara. The terminology and boundaries of these categories, particularly Vesara, are debated in architectural scholarship. The project should therefore use carefully defined labels and explicitly document the classification scheme adopted for the dataset.

> A suitable dataset label can be “Deccan / traditionally classified Vesara” when the underlying sources support that terminology. The model should report a classification under the project's operational taxonomy rather than presenting it as an absolute historical truth.

12. Proposed Solution

```
Temple Image
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■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓ ↓
Whole-image Encoder Component Detector
■ ■
↓ ↓
Visual Features Architectural Features
■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓
Feature Fusion
↓
Style Classifier
↓
Explainability Layer
↓
IKS Knowledge Layer
```

```
↓
Architectural Explanation
```

13. Dataset Strategy

The dataset should combine publicly available, appropriately licensed temple images with structured metadata. Potential sources include Wikimedia Commons, institutional digital heritage archives and public-domain or openly licensed collections. Every image should retain provenance and license information.

Recommended metadata fields:

- image_id
- temple_id
- temple_name
- location
- state/region
- architectural_tradition
- historical_period
- dynasty where supported by the source
- architectural_feature_labels
- source
- license
- image_type

14. Temple-Disjoint Evaluation

This is one of the most important methodological requirements. Do not randomly split photographs if multiple photographs of the same temple exist. Otherwise, the model may memorize monument-specific visual characteristics.

```
Training:
Temple A, Temple B, Temple C, Temple D

Testing:
Temple E, Temple F, Temple G

Rule:
No temple appears in both training and test sets.
```

A temple-disjoint protocol is also consistent with recent research methodology in Indian temple classification and provides a more realistic test of generalization.

15. Architectural Component Dataset

The second layer of the dataset should annotate architectural components. Depending on resources, annotations can be bounding boxes, segmentation masks or structured image-level labels.

- ■ikhara
- Vim■na
- Gopuram

- Ma■■apa
- Pillars
- Jagati/base
- Amalaka
- Other project-specific components supported by the literature

16. Baseline Models

16.1 Conventional CNN Baseline

```
Image
↓
EfficientNet-B0
↓
Global Average Pooling
↓
Fully Connected Layer
↓
Architectural Class
```

EfficientNet-B0 is a useful baseline because recent Indian temple classification research has evaluated it directly.

16.2 Lightweight Baseline

```
Image
↓
MobileNetV3
↓
Classifier
↓
Architectural Class
```

MobileNetV3 can provide an accuracy-efficiency baseline and a pathway toward future mobile or edge deployment.

17. Proposed Feature-Aware Model

```
Temple Image
■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓ ↓
Whole-image CNN Component Detector
■ ■
↓ ↓
Global Visual Features Architectural Features
■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓
Feature Fusion
↓
Classification Head
↓
Architectural Style
↓
Explainability + IKS
```

The proposed model combines global visual representations with explicit architectural-component information. The exact fusion mechanism can be concatenation followed by fully connected layers, attention-based fusion, or another experimentally justified approach.

18. Explainable AI

18.1 Grad-CAM

Grad-CAM can visualize the image regions contributing to a CNN prediction. In this project, it should be evaluated against annotated architectural regions rather than presented only as a qualitative heatmap.

18.2 Feature Attribution

SHAP or another suitable attribution method can be used for interpretable feature contribution analysis, particularly for structured architectural features.

18.3 Component-Level Explanation

The target explanation is not merely “upper-left region was important.” Instead, the system should connect visual evidence to a detected architectural component such as a shikhara, vimana, gopuram or mandapa.

19. Knowledge-Grounded Explanation

```
Visual Evidence
+
Detected Architectural Components
+
Structured IKS Knowledge
↓
Evidence-grounded explanation
↓
Human-readable architectural report
```

An optional LLM can convert structured evidence into natural language, but it should not independently decide the architectural class. The classification should come from the validated computer-vision pipeline and knowledge layer.

20. Full System Architecture

```
USER
■
↓
Upload Temple Image
■
↓
Image Preprocessing
■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓ ↓
Whole-image CNN Architectural Detector
■ ■
↓ ↓
Visual Features Component Features
■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓
Feature Fusion
↓
Architectural Classifier
■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓ ↓
Style Prediction Component Detection
■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓
Explainable AI
↓
IKS Knowledge Base
↓
Architectural Report
```

21. User Interface

A practical prototype can be implemented as a React/TypeScript frontend with a FastAPI or Flask backend.

- Upload temple image
- Display predicted architectural tradition and confidence
- Display detected architectural components
- Show bounding boxes/segmentation masks
- Show Grad-CAM or other explanation visualization
- Display evidence-grounded architectural explanation
- Display relevant source/provenance information

22. Evaluation Metrics

Category	Metrics
Classification	Accuracy, Precision, Recall, Macro F1, Weighted F1
Per-class performance	Confusion matrix, per-class Precision/Recall/F1
Explainability	Pointing Game, explanation-region IoU, occlusion/faithfulness tests
Component detection	mAP, precision, recall, IoU; segmentation IoU/Dice if masks are used
Robustness	Performance under crops, lighting changes, occlusion and background perturbations
Efficiency	Inference time, parameter count and model size where relevant

23. Explanation Evaluation

A strong paper should evaluate whether the explanation corresponds to actual architectural evidence. One practical method is to compare explanation regions against human-annotated architectural components using Intersection over Union (IoU). Another is to mask the model's most important region and measure whether prediction confidence changes substantially.

IoU: $IoU = Area(A \cap B) / Area(A \cup B)$, where A is the explanation region and B is the annotated architectural component.

24. Ablation Study

Run at least four experiments:

Experiment	Input/architecture
A	CNN baseline only
B	CNN + architectural components
C	CNN + components + structured knowledge
D	Full proposed model + explainability/evidence alignment

Compare classification performance, generalization and explanation quality. Do not invent numerical results before experiments are conducted.

25. Robustness Experiments

- Different lighting conditions
- Different camera viewpoints
- Partial occlusion
- Tourists/people in foreground
- Trees and sky in background
- Low-resolution images
- Cropped architectural regions
- Background removal or masking

26. Research Novelty

- **Feature-aware classification:** combines whole-image features with explicit architectural components.
- **Architecture-specific XAI:** explanations are tied to meaningful architectural structures.
- **Knowledge-grounded interpretation:** visual evidence is mapped to a structured IKS architectural knowledge layer.
- **Temple-disjoint evaluation:** tests generalization to unseen monuments.
- **Potential multimodal extension:** combines visual information with textual architectural descriptions.

27. Optional Multimodal Extension

```
Image ■■→ Vision Encoder ■■■
■■■→ Multimodal Fusion → Classification
Text ■■■→ Text Encoder ■■■■■■■
```

The text input may contain a sourced architectural description. A multimodal model can test whether visual and textual evidence together improve recognition or explanation.

28. Optional Knowledge Graph

```
Temple
■
■■■ Location
■■■ Period
■■■ Dynasty
■■■ Architectural Tradition
■■■ ■ikhara Type
■■■ Vim■na
■■■ Ma■■apa
■■■ Gopuram
■■■ Source
```

A knowledge graph can provide provenance-aware relationships among temples, architectural components, regions, periods and sources.

29. Optional 3D Extension

If the group has access to photogrammetry, LiDAR, drone imagery or smartphone depth scanning, the project can be extended toward 3D architectural-component segmentation and digital reconstruction. This should remain an extension rather than a requirement for the initial project because it substantially increases data and implementation complexity.

30. Technology Stack

Layer	Recommended technologies
Programming	Python, TypeScript
Deep learning	PyTorch, torchvision
Computer vision	OpenCV, YOLO, SAM/SAM2 where justified
Explainability	Grad-CAM, SHAP, Integrated Gradients / occlusion analysis
Data	Pandas, NumPy, scikit-learn
Backend	FastAPI or Flask
Frontend	React, TypeScript, Tailwind CSS
Database	PostgreSQL/Supabase; Neo4j for knowledge graph if required
Deployment	Docker; Cloud Run/AWS/Azure as appropriate

31. Development Timeline

Weeks	Work
1	IKS and temple-architecture literature study; define taxonomy
2	Literature survey and research-gap analysis
3–4	Dataset collection, cleaning and metadata creation
5	EfficientNet/MobileNet baseline
6–7	Architectural component detection/segmentation
8	Feature-fusion model
9	Grad-CAM and explanation evaluation
10	Knowledge layer and provenance
11	Ablation, robustness and generalization experiments
12	React + FastAPI prototype
13–14	Research paper, documentation and presentation

32. Team Division

Member	Responsibilities
Member 1	IKS literature, architectural taxonomy, source verification, metadata
Member 2	Dataset construction, computer vision, component detection
Member 3	Feature fusion, XAI, experiments, ablation and evaluation
Member 4	Backend, frontend, database, visualization and deployment

33. Expected Results

- A model capable of classifying selected temple architectural traditions under a clearly defined taxonomy.

- Explicit identification of architectural components.
- Visual and feature-level explanations for predictions.
- Quantitative comparison with conventional image-level baselines.
- Temple-disjoint evaluation demonstrating generalization to unseen monuments.
- A searchable/provenance-aware architectural knowledge layer.
- A working web prototype for demonstration and educational use.

34. Expected Deliverables

- Annotated Indian temple image dataset with provenance and licensing information.
- Feature-aware temple architecture classifier.
- Architectural component detector/segmenter.
- Explainable AI module.
- IKS architectural knowledge base or knowledge graph.
- Web-based demonstration platform.
- Research paper manuscript.
- Potential software copyright or other IP assessment if the final implementation contains sufficiently original work.

35. Publication Strategy

The project should be positioned as a research contribution rather than as a generic application. Suitable venues depend on the final methodology and results and should be verified at submission time. Relevant areas include computer vision, digital cultural heritage, heritage informatics, AI and architectural computing.

The goal should be a legitimate peer-reviewed venue whose scope matches the actual contribution. “Scopus” is an indexing database and should not be treated as a quality guarantee by itself.

36. Suggested Five-Paper Literature Starting Set

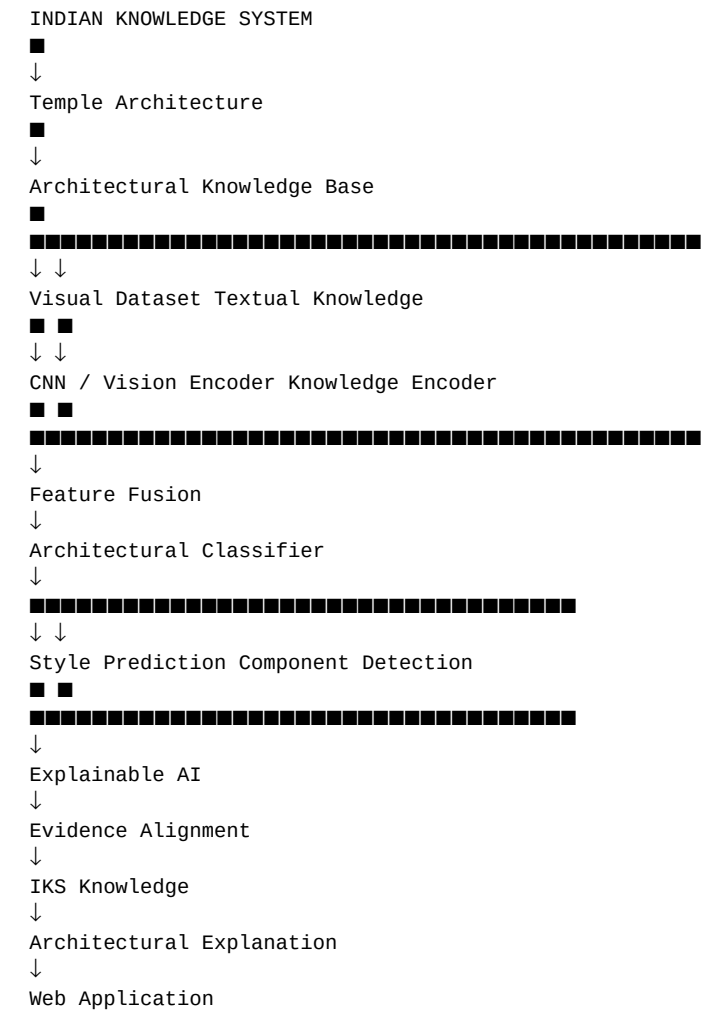
Research area	Why it matters to this project
Indian temple architectural style recognition and interpretation (2026)	Direct baseline: image classification, lightweight CNNs and Grad-CAM.
3D semantic segmentation / reconstruction of Indian temples / heritage architecture (2026)	Shows how to move from image-level analysis to component-level analysis and digital reconstruction.
3DITA — 3D benchmark for Nagara-style Indian temple architecture	Provides a heritage point-cloud dataset direction.
X-NeSyL / MonuMAI cultural-heritage work	Highly relevant to combining deep learning with expert knowledge graphs and explainability.
Digital heritage documentation using imaging/scanning technologies	Provides context for computational preservation and documentation.

37. Phase-1 250-Word Abstract

Indian temple architecture represents a significant component of the Indian Knowledge System, incorporating structured principles of spatial organization, geometry, construction, ornamentation and architectural design. Traditional architectural scholarship describes diverse temple traditions including N■gara, Dr■vi■a and Deccan architectural forms, along with numerous regional and historical variations. However, identification and documentation of architectural styles from visual data still relies substantially on human expertise. This project proposes an Explainable

Multimodal Artificial Intelligence framework for automated recognition and interpretation of Indian temple architectural heritage. The proposed system will use computer vision and deep learning to classify temple architectural traditions while explicitly identifying meaningful architectural components such as shikhara, vimana, gopuram, mandapa, pillars and structural bases. Unlike conventional image-level classification systems, the proposed approach will combine whole-image visual representations with component-level architectural features and structured Indian architectural knowledge. Explainable AI techniques such as Grad-CAM and feature attribution will be employed to identify the visual regions and architectural components influencing model predictions. A temple-disjoint evaluation strategy will be adopted to measure generalization to previously unseen temples and reduce the possibility of monument-specific memorization. The project will further investigate whether incorporating architectural features improves classification performance and explanation quality compared with conventional deep-learning baselines. The expected outcome is a prototype platform capable of analyzing temple images, identifying architectural characteristics, providing interpretable predictions and presenting relevant knowledge about the detected architectural tradition. The project aims to contribute toward AI-assisted digital preservation, architectural education, heritage documentation and computational research in Indian Knowledge Systems.

38. Recommended Final Research Framework



39. Conclusion

The proposed project provides a strong intersection between Indian Knowledge Systems and modern computer science. Its strongest research direction is not simply automated temple-style classification, because that problem has already received recent research attention. The project should instead investigate whether explicit architectural components, explainability and structured IKS knowledge can make AI-based heritage analysis more reliable, interpretable and useful for previously unseen temples.

The recommended contribution is therefore: **classification + architectural component detection + explanation alignment + IKS knowledge grounding + temple-disjoint evaluation**. This combination provides a clear experimental methodology, measurable research questions, a working software prototype and a plausible path toward a peer-reviewed research paper.

40. Selected References / Starting Sources

- Indian temple architectural style recognition and interpretation using deep learning. Digital Applications in Archaeology and Cultural Heritage, 2026. Directly relevant baseline on temple image classification and Grad-CAM.
- Deep learning-based semantic segmentation / 3D reconstruction research for Indian temple heritage. Measurement, 2026. Relevant to 3D component-level heritage analysis.
- 3DITA — A 3D benchmark dataset for Nagara-style Indian temple architecture. ISPRS Archives, 2025.
- X-NeSyL / MonuMAI cultural-heritage research on combining deep learning representations with expert knowledge graphs for explainability.
- Digital heritage documentation research using scanning, photogrammetry and imaging technologies for Indian monuments.

41. Important Research-Design Note

Do not report accuracy, F1, IoU or other numerical results until the experiments have actually been performed. Do not claim that a model “understands” Indian architecture merely because it produces a plausible label. The paper should distinguish between model predictions, source-backed architectural knowledge and human interpretation. This separation will substantially improve the academic credibility of the project.